

CCNA — Assessment File M4-A1
Module 4 — Routing and Switching

A Section A — Concept Application

S1

SCENARIO

You are the network administrator for a small startup with only two routers connecting two offices, and the network topology rarely changes.

Question: Explain the operational differences between static and dynamic routing, and justify why configuring static routes would be the most efficient choice in this scenario.

Answer:

Static vs Dynamic Routing

- Static routing: we manually type each route into the router's routing table (`ip route <destination> <mask> <next-hop>`). The router never recalculates anything — it just uses the route it is told.
- Dynamic routing: routers run a protocol (RIP, OSPF, EIGRP, etc.) that automatically discovers neighbors, exchanges routing information, and recalculates paths when the topology changes.

Key operational differences:

	Static	Dynamic
Configuration effort	Manual, per route	Automatic once protocol is enabled
CPU/memory	None	Some (routing updates, calculations)
Reacts to link failure	No — needs manual fix	Yes — recalculates path automatically
Security	More predictable, no advertised routes	Routes advertised, prone to attack
Scalability	Poor for large/changing topologies	Good

Here static is best because two routers, two offices, one link between them, topology rarely changes. There's only one possible path anyway, so there's nothing for a dynamic protocol to "discover" or optimize, it just adds CPU overhead, bandwidth used for periodic updates, and a bit of convergence delay for zero benefit. A single static route on each router accomplishes the same result instantly and predictably.

S2

SCENARIO

You are expanding the startup's network to five branch routers, and manually maintaining routes has become error-prone.

Question: Compare RIP and OSPF as dynamic routing protocol options for this network, and recommend which one is better suited as the network grows, justifying your choice based on convergence time and scalability.

Answer:

Distance Vector Protocols (RIP)

- Each router only knows what its directly connected neighbors tell it — the “distance” (metric, e.g. hop count) and “direction” (which neighbor leads there).
- Routers periodically share their entire routing table with directly connected neighbors (every 30 seconds for RIP).
- No router ever builds a full picture of the network topology; each one simply trusts what its neighbors report.
- Simple to configure, but converges more slowly and is more prone to routing loops, requiring loop-prevention mechanisms such as split horizon, poison reverse, and hold-down timers.

Link-State Protocols (OSPF)

- Each router builds a complete, identical map of the entire network topology (within its area), not just what its neighbors say.
- Routers flood Link-State Advertisements (LSAs) describing their own directly connected links to every router in the area — not just to direct neighbors.
- Every router independently runs the same algorithm (Dijkstra's Shortest Path First) over that shared map to calculate its own best path to every destination.
- Converges much faster and avoids loops more reliably than distance vector, but uses more CPU and memory, and is more complex to configure and troubleshoot.

Recommendation: OSPF — as the network grows past a handful of routers, RIP's hop-count ceiling and slow convergence become real liabilities, while OSPF is built for exactly this kind of scaling and gives faster recovery when a branch link fails.

S3

SCENARIO

You are troubleshooting two newly connected startup devices attempting to communicate for the first time through the same switch.

Question: Explain how the switch populates its MAC address table and forwards frames between the two devices, including what happens before the table is populated.

ANS: A switch initially contains not MAC address, it builds its MAC address table dynamically by learning from the source MAC address of every frame that arrives on a port.

Phase 1: Before the Table is Populated (The Initial Frame)

- **Table State:** The MAC address table is completely empty upon startup.
- **Device A Transmits:** Device A sends an Ethernet frame destined for Device B.
- **Source Learning:** The switch receives the frame, reads Device A's source MAC address, and registers it in the table alongside the incoming physical port number.
- **Unknown Unicast Flooding:** The switch reads the destination MAC address for Device B. Because Device B is not yet in the table, the switch performs **unknown unicast flooding**, duplicating and broadcasting the frame out of every active port except the port where Device A is connected.

Phase 2: The Return Traffic and Table Completion

- **Device B Receives:** Device B identifies its own MAC address in the frame and processes it. Other devices on the switch will drop the frame.
- **Device B Responds:** Device B sends a reply frame back to Device A.

- **Second Source Learning:** The switch receives the frame from Device B, reads its source MAC address, and maps Device B's MAC address to its connected physical port in the table.
- **Known Unicast Forwarding:** The switch looks up Device A's destination MAC address. Because Device A is now registered in the table from Phase 1, the switch forwards the frame directly out of Device A's port without flooding. Future communications between both devices are handled via direct, point-to-point forwarding.

S4

SCENARIO

You are designing the network for a startup with a Marketing team and an Engineering team that must be logically separated for security, even though both teams' devices connect to the same physical switch.

Question: Explain the purpose of VLAN segmentation, and describe how an 802.1Q trunk link allows traffic for multiple VLANs to travel between the switch and the router over a single physical connection.

Answer:

A VLAN divides a physical switch into multiple logical switches, each forming its own broadcast domain.

- Security: devices in one VLAN cannot connect with another device in a different VLAN at layer 2.
- Reduced Broadcast: Broadcast stays inside own VLAN, which improves performance.
- Flexibility: Groups are organized by function rather than by physical location, with no need for separate switches per team.

802.1Q trunk link

- The end devices connected to the switch ports are access ports, each belonging to a single VLAN. The link between the switch and router however must carry traffic from both VLANs. So instead of one cable per VLAN, we use a single link configured as an 802.1Q trunk.
- When a frame leaves the switch over the trunk, the switch inserts a 4-byte **802.1Q tag** into the Ethernet header. This tag includes the **VLAN ID**.
- The router uses **sub interfaces** (router-on-a-stick), for example G0/0.10 for VLAN 10 and G0/0.20 for VLAN 20. Each is set with encapsulation dot1Q <vlan-id> and given an IP address that serves as the default gateway for that VLAN.
- The router routes traffic between VLANs, then re-tags the frame with the destination VLAN ID and sends it back over the same trunk. The switch removes the tag and delivers the frame out the correct access port.
- Frames from the **native VLAN** (VLAN 1 by default) are sent untagged, so the native VLAN must match on both ends of the trunk.

Multilayer Switch0

Physical | Config | CLI | Attributes

IOS Command Line

Output Modifiers

```

<cr>
Switch#show interfaces trunk ?
  | Output Modifiers
<cr>
Switch#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/1     1-1005

Port      Vlans allowed and active in management domain
Fa0/1     1,10,20,30

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1,10,20,30

Switch#show vlan brief

```

VLAN	Name	Status	Ports
1	default	active	Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10	VLAN0010	active	Fa0/2
20	VLAN0020	active	Fa0/3
30	VLAN0030	active	Fa0/4
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```

Switch#

```

S5

SCENARIO

You are configuring connectivity so that the Marketing VLAN and Engineering VLAN, which currently cannot communicate, can reach each other through the startup's single router.

Question: Explain why inter-VLAN routing is required here, and justify the specific advantages of a Router-on-a-Stick configuration over adding a second physical router interface.

Answer:

Why inter-VLAN routing is required:

Each VLAN is its own broadcast domain and its own IP subnet. In your lab, VLAN 10 uses 192.168.10.0/24 and VLAN 20 uses 192.168.20.0/24. A switch forwards frames only within a VLAN at Layer 2, so a PC in VLAN 10

cannot reach a PC in VLAN 20 even when both are on the same physical switch. Traffic between different subnets needs a Layer 3 device, so a router must receive the packet from one VLAN, look up the destination network in its routing table, and forward it into the other VLAN. This is also where security policy can be enforced, because the router can apply ACLs to control what Marketing and Engineering may reach.

Advantages of Router-on-a-Stick over a second physical interface:

1. **Fewer interfaces and cables.** The alternative uses one router interface and one switch port per VLAN. Router-on-a-Stick uses a single physical link (Router Fa0/0 to Switch Fa0/24) carried as an 802.1Q trunk, which saves ports on both devices.
2. **Lower cost.** The startup can use its single router without buying extra devices.
3. **Scalability.** Adding a new VLAN only takes a new subinterface (for example Fa0/0.30) and no new cabling. A physical-interface design runs out of ports quickly.
4. **Simpler management.** All the routing configuration sits on one physical interface, which makes the layout cleaner and easier to troubleshoot.
5. **Same routing result.** Each subinterface has its own IP address, which acts as that VLAN's default gateway, so the router routes between VLANs just as it would with separate interfaces.

S6

SCENARIO

You are setting up a wireless access point for the startup's open-plan office, where unauthorized devices have previously connected to the network.

Question: Explain the wireless security and MAC filtering options available, and justify which configuration approach best prevents unauthorized access while still allowing convenient onboarding of new employee devices.

Answer:

Wireless security options:

Method	How it works	Strength
Open	No authentication, no encryption	None — anyone can connect and read traffic
WEP	Static key, weak RC4 encryption	Broken; crackable in minutes, never use it
WPA	TKIP (temporal key integrity protocol) encryption	Deprecated, has known weaknesses
WPA2-Personal (PSK)	Shared passphrase, AES-CCMP encryption. AES – Advanced encryption standard) PSK -Pre-shared key	Good, but the same password is known to everyone, and it can't be revoked per user
WPA2/WPA3-Enterprise (802.1X)	Each user authenticates individually against a RADIUS server, typically with their own credentials or certificate	Strongest — unique per-user credentials, centrally revocable, no shared secret
WPA3-Personal (SAE)	Replaces PSK's key exchange with SAE, resists offline dictionary attacks (SAE -Simultaneous Authentication of Equals)	Good upgrade over WPA2-PSK if all devices support it

MAC filtering:

The Access Points maintains a list of allowed (or denied) MAC addresses and blocks association from anything not on the list.

- Limitation: When devices talk over Wi-Fi, the actual data (like your passwords or messages) are encrypted by WPA2/WPA3. However, the MAC addresses are not encrypted. "Spoofing" means faking an identity. Because it is incredibly easy to capture a wifi signal and change a device's reported MAC address using software, an intruder doesn't need to hack your hardware, instead he can spoof the allowed MAC address in the network and gain access.
- Limitation: Every new employee device (phone, laptop, tablet) needs to be manually added, which is an administrative burden in an open-plan office with frequent on boarding.

Recommended approach for this scenario:

WPA2-Enterprise (802.1X with RADIUS), optionally paired with MAC filtering as a secondary layer.

Justification:

1. Each employee authenticates with their own username/password or certificate. A former employee or unauthorized device can't connect just by learning a shared Pre-Shared Key or a spoofed MAC
2. New employees are added once to the RADIUS server, and their devices connect using their normal login. No need to hand out a shared password or manually collect every device's MAC address.
3. Revocation without disruption. If a device is lost or an employee leaves, you disable that one account. But Wwth WPA2-PSK, you'd have to change the passphrase and redistribute it to everyone.
4. Accountability. Logs show which user connected when, useful for troubleshooting and investigating the unauthorized-access history mentioned in the scenario.
5. MAC filtering as a supplementary control can still be layered on for a small, tightly managed set of fixed devices (like a lobby printer or IoT device that can't do 802.1X), but it should not be the primary control given how easily it's spoofed.

B Section B — Practical Configuration Tasks

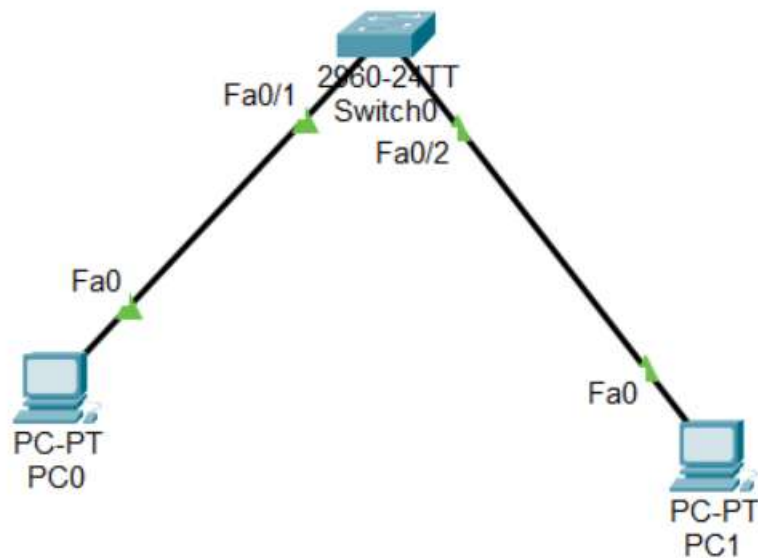
Complete all tasks in Cisco Packet Tracer. Each task targets one specific concept or feature from the module. Follow the requirements listed under each task.

Task 1: Configure Management and Staff VLANs

Set up two VLANs on a Cisco switch to logically separate startup departments. (Foundational — single concept, direct configuration.)

- Create VLAN 10 named Management and VLAN 20 named Staff on the switch.
- Assign specific access ports to each VLAN.
- Verify the VLAN assignments using show vlan brief.

Answer:



```
Switch0
Physical | Config | CLI | Attributes |
IOS Command Line Interface
Switch(config-if)#do wr
Building configuration...
[OK]
Switch(config-if)#exit
Switch(config)#int fa0/2
Switch(config-if)#switchport access vlan 20
Switch(config-if)#do wr
Building configuration...
[OK]
Switch(config-if)#exit
Switch(config)#end
Switch#
*SYS-5-CONFIG_I: Configured from console by console

Switch#show vlan brief

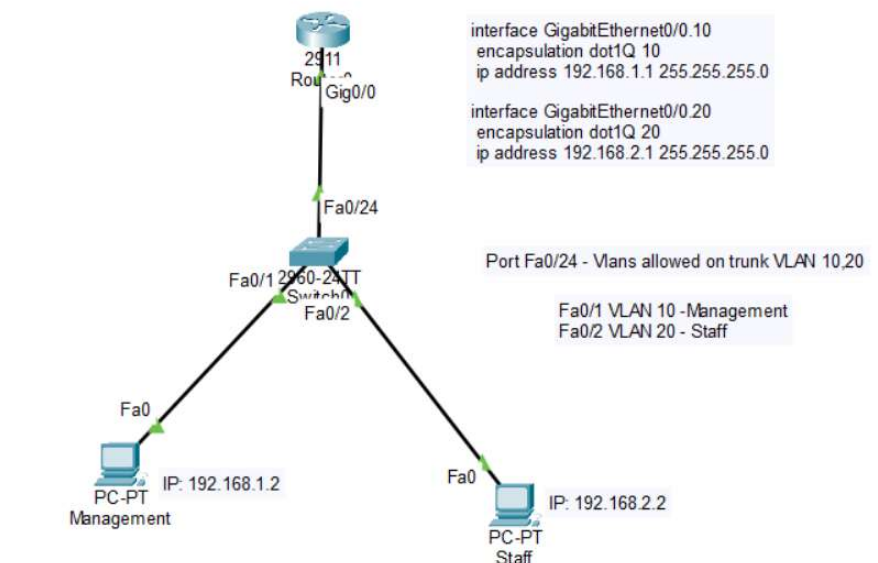
VLAN Name                Status    Ports
-----
1    default                active    Fa0/3, Fa0/4, Fa0/5, Fa0/6,
Fa0/7, Fa0/8, Fa0/9, Fa0/10,
Fa0/11, Fa0/12, Fa0/13,
Fa0/14
Fa0/18
Fa0/22
Fa0/23, Fa0/24, Gig0/1,
Gig0/2
10   Management              active    Fa0/1
20   Staff                  active    Fa0/2
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default        active
1005 trnet-default          active
Switch#
```

Task 2: Configure an 802.1Q Trunk Link

Establish a trunk connection so traffic for multiple VLANs can pass between the switch and router. (Applied — one concept combined with port-mode configuration.)

- Configure the switch port connecting to the router as an 802.1Q trunk.
- Restrict the trunk to allow only VLAN 10 and VLAN 20.
- Verify trunk status using show interfaces trunk.
- Confirm there is no native VLAN mismatch between the switch and router.

Answer:



Switch0

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
<cr>
Switch#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/24    on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/24    10,20

Port      Vlans allowed and active in management domain
Fa0/24    10,20

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/24    20

Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default              active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                           Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                           Fa0/11, Fa0/12, Fa0/13,
                                           Fa0/14
                                           Fa0/15, Fa0/16, Fa0/17,
                                           Fa0/18
                                           Fa0/19, Fa0/20, Fa0/21,
                                           Fa0/22
10   Management            active    Fa0/23, Gig0/1, Gig0/2
20   Staff                 active    Fa0/1
                                           Fa0/2
1002 fddi-default          active
1003 token-ring-default  active
1004 fddinet-default      active
1005 trnet-default        active
Switch#
```

Copy Paste

Task 3: Configure Router-on-a-Stick for Inter-VLAN Routing

Enable communication between VLAN 10 and VLAN 20 through router sub-interfaces. (Intermediate — two concepts used together: sub-interfacing and routing.)

- Create two sub-interfaces on the router's trunk-connected interface, one per VLAN.
- Assign correct IP addressing and 802.1Q encapsulation with the matching VLAN ID on each sub-interface.
- Enable routing so hosts in VLAN 10 and VLAN 20 can reach each other.
- Verify connectivity with a successful ping between hosts in different VLANs.

Answer:

```
Router0
Physical | Config | CLI | Attributes |
IOS Command Line Interface
!
!
!
!
!
!
!
interface GigabitEthernet0/0
  no ip address
  duplex auto
  speed auto
!
interface GigabitEthernet0/0.10
  encapsulation dot1Q 10
  ip address 192.168.1.1 255.255.255.0
!
interface GigabitEthernet0/0.20
  encapsulation dot1Q 20
  ip address 192.168.2.1 255.255.255.0
!
interface GigabitEthernet0/1
  no ip address
  duplex auto
  speed auto
  shutdown
!
interface GigabitEthernet0/2
  no ip address
  --More--
```

Staff

PhysicalConfigDesktopProgrammingAttributes

Command Prompt

X

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=127
Reply from 192.168.1.2: bytes=32 time<0ms TTL=127
Reply from 192.168.1.2: bytes=32 time<1ms TTL=127
Reply from 192.168.1.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.1.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>

Management

PhysicalConfigDesktopProgrammingAttributes

Command Prompt

X

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.2.2:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms

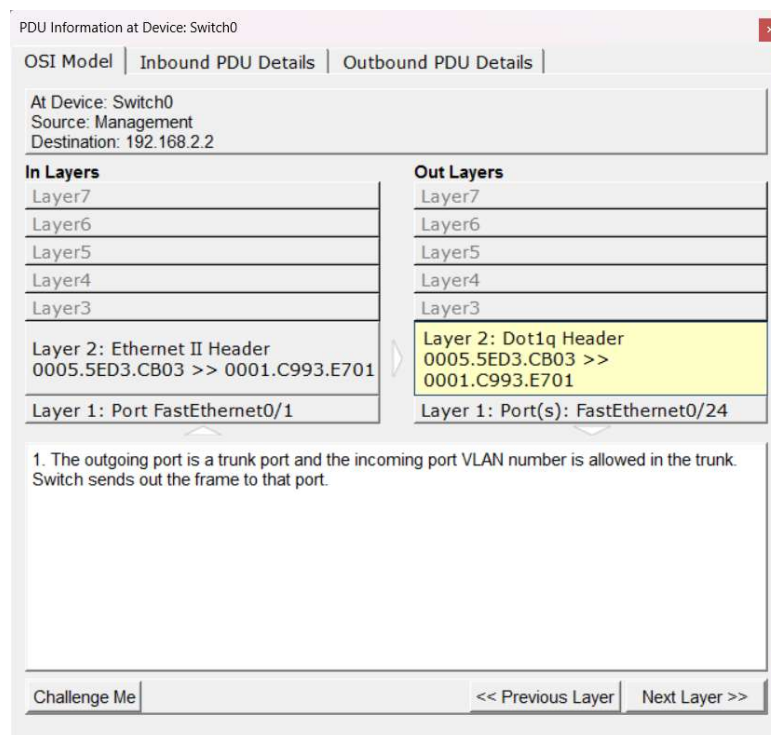
C:\>

Task 4: Capture and Analyze Inter-VLAN Traffic

Use Wireshark to observe how frames change as they cross from switch to router and back. (Integrative — multi-step flow combining VLANs, trunking, and routing.)

- Capture ICMP traffic while pinging between a VLAN 10 host and a VLAN 20 host.
- Identify the specific frames that cross the router via the trunk link.
- Document the source/destination MAC and IP addresses observed at each hop.
- Explain how the frame headers change as traffic passes from switch to router and back.

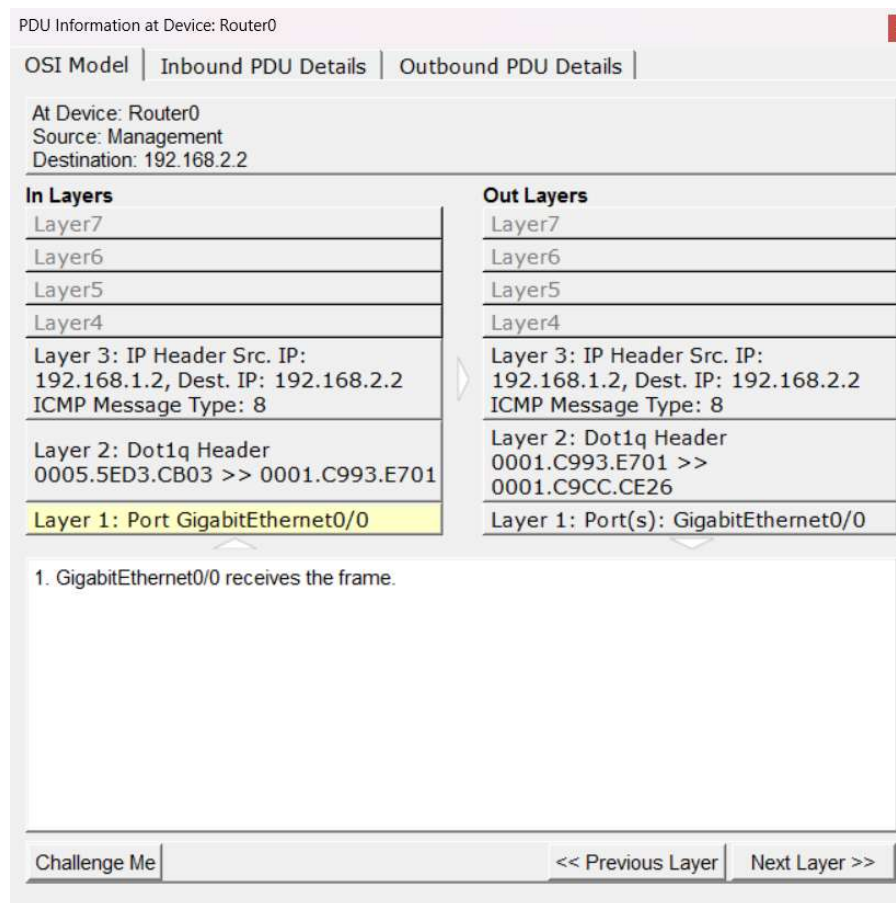
Answer:



Hop 1: Switch0 (Management PC → Router, across the trunk)

	Inbound (Fa0/1, access port)	Outbound (Fa0/24, trunk port)
Header type	Ethernet II	802.1Q
Src MAC	0005.5ED3.CB03 (Management PC)	0005.5ED3.CB03 (unchanged)
Dst MAC	0001.C993.E701 (Router Gi0/0.10)	0001.C993.E701 (unchanged)

What happened: the switch received a plain, untagged Ethernet II frame on the access port and simply **added a VLAN tag** (VLAN 10) before forwarding it out the trunk. No MAC addresses changed — the switch never rewrites addresses, it only adds or strips 802.1Q tags.



Hop 2: Router0 (receiving from trunk, routing, sending back out)

Inbound (Gi0/0, tagged VLAN 10)		Outbound (Gi0/0, tagged VLAN 20)	
Src IP	192.168.1.2	192.168.1.2 (unchanged)	
Dst IP	192.168.2.2	192.168.2.2 (unchanged)	
Src MAC	—	0001.C993.E701 (Router's own MAC, now sourced from Gi0/0.20)	
Dst MAC	0001.C993.E701 (Router — frame's destination)	0001.C9CC.CE26 (Staff PC — new destination)	
VLAN tag	10 (dot1Q, matches Gi0/0.10)	20 (dot1Q, matches Gi0/0.20)	

What happens at the router:

1. Received the frame on Gi0/0.10 (VLAN 10) and removed the Ethernet/802.1Q header entirely, unmasking the IP packet.
2. It reads the destination IP (192.168.2.2), checks with its routing table, and matches it to the directly connected 192.168.2.0/24 network on Gi0/0.20.
3. **Built a brand-new Layer 2 header** for the outbound frame: source MAC is the router's own interface MAC (0001.C993.E701 — subinterfaces of Gi0/0 share one MAC address), and destination MAC is now **0001.C9CC.CE26**, the Staff PC's MAC address, which the router already had saved in ARP.

4. Tagged the new frame with **VLAN 20** instead of VLAN 10, since it's exiting via Gi0/0.20.
5. The IP header (source 192.168.1.2, destination 192.168.2.2) stayed **completely unchanged**.

C Section C — Mini Capstone Project

Mini Project: TechLaunch Startup Network Build & Fault Diagnosis

Objective:

Build a complete two-VLAN network for a startup, then diagnose and resolve a Router-on-a-Stick failure following a power outage — combining VLAN segmentation, trunking, inter-VLAN routing, and structured fault-finding into one exercise.

Your project must:

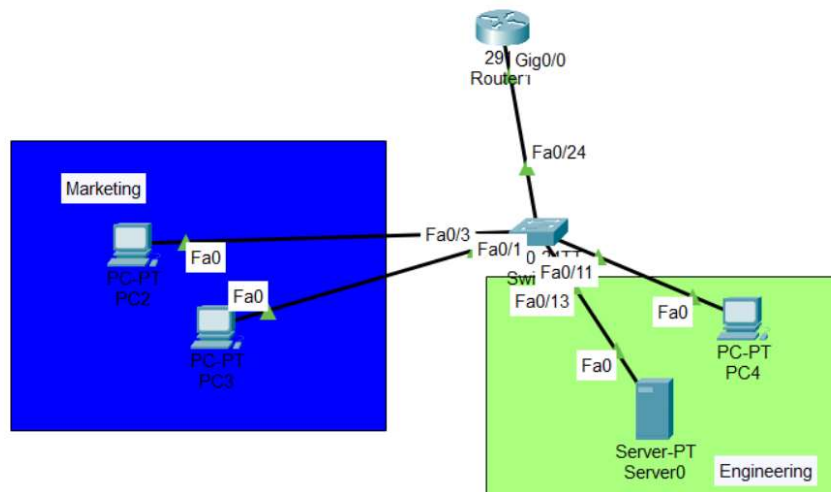
- Design and configure VLAN 10 (Marketing) and VLAN 20 (Engineering) on the switch, assigning appropriate ports to each department.
- Configure an 802.1Q trunk link between the switch and router carrying both VLANs.
- Configure Router-on-a-Stick sub-interfaces so Marketing and Engineering can reach each other after the router's post-outage reboot.
- Given that Marketing cannot reach Engineering's file server despite being able to ping devices within its own VLAN, use a documented ping-based process to isolate whether the fault is at Layer 2 or Layer 3.
- Produce a short written report listing the exact CLI verification commands used and the root cause identified.

Answer:

Network Design

VLAN	Department	Subnet	Gateway	Switch ports
10	Marketing	192.168.20.0/28 (255.255.255.240)	192.168.20.1	Fa0/1 - Fa0/10
20	Engineering	192.168.20.16/28 (255.255.255.240)	192.168.20.17	Fa0/11 - Fa0/20

- **Trunk:** switch port Fa0/24 to router GigabitEthernet0/0 is designated as trunk port, carrying VLANs 10 and 20.
- **Router-on-a-Stick:** sub-interface Gi0/0.10 serves VLAN 10 and Gi0/0.20 serves VLAN 20.
- **DHCP:** the router hands out addresses from two pools, one per VLAN.
- **Engineering file server (Server0):** static address 192.168.20.19, mask 255.255.255.240, gateway 192.168.20.17.



Router (sub-interfaces and DHCP)

Sub-interfaces on Gi0/0 was created and use 802.1Q encapsulation and each holds the default gateway address of its VLAN.

```

Router1
Physical | Config | CLI | Attributes |
IOS Command Line Interface

!
!
spanning-tree mode pvst
!
!
!
!
!
!
!
interface GigabitEthernet0/0
no ip address
duplex auto
speed auto
!
interface GigabitEthernet0/0.10
encapsulation dot1Q 10
ip address 192.168.20.1 255.255.255.240
!
interface GigabitEthernet0/0.20
encapsulation dot1Q 20
ip address 192.168.20.17 255.255.255.240
!
interface GigabitEthernet0/1
no ip address
duplex auto
speed auto
shutdown
!
interface GigabitEthernet0/2
no ip address
duplex auto

```

Figure 1: Router1 running-config - sub-interfaces Gi0/0.10 and Gi0/0.20

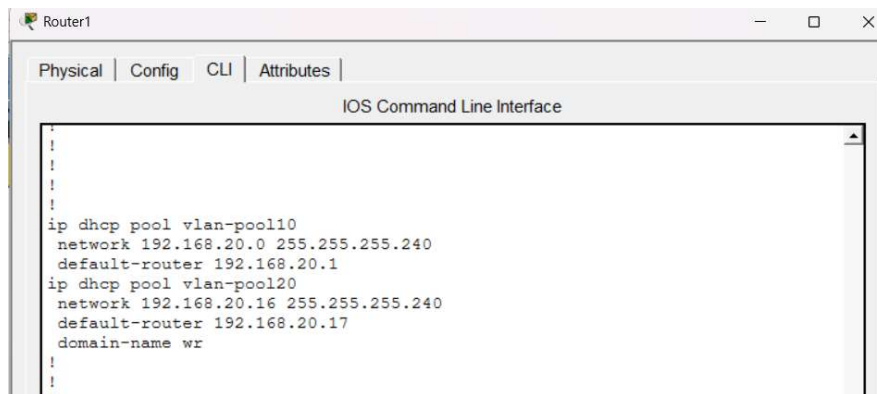


Figure 2: Router1 running-config - DHCP pools vlan-pool10 and vlan-pool20

Working Network

PC2 (VLAN 10) pinged PC4 (192.168.20.18) in VLAN 20. Three of four replies returned with TTL=127. A PC starts at TTL 128 and each router hop subtracts 1, so TTL 127 proves the packet crossed the router. The first "Request timed out" is normal: the first packet waits while ARP resolves the address.

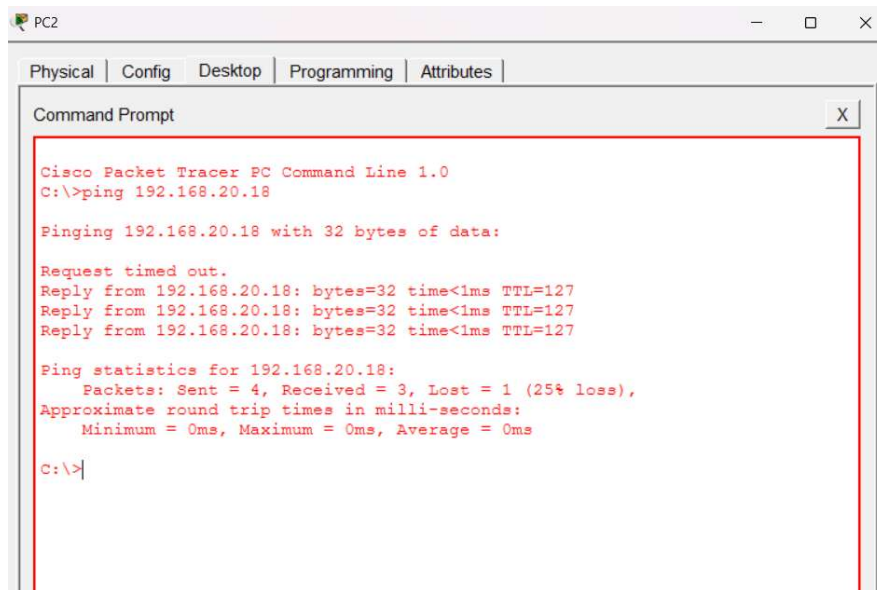


Figure 3: PC2 pings 192.168.20.18 - inter-VLAN routing works (TTL=127)

Fault Diagnosis

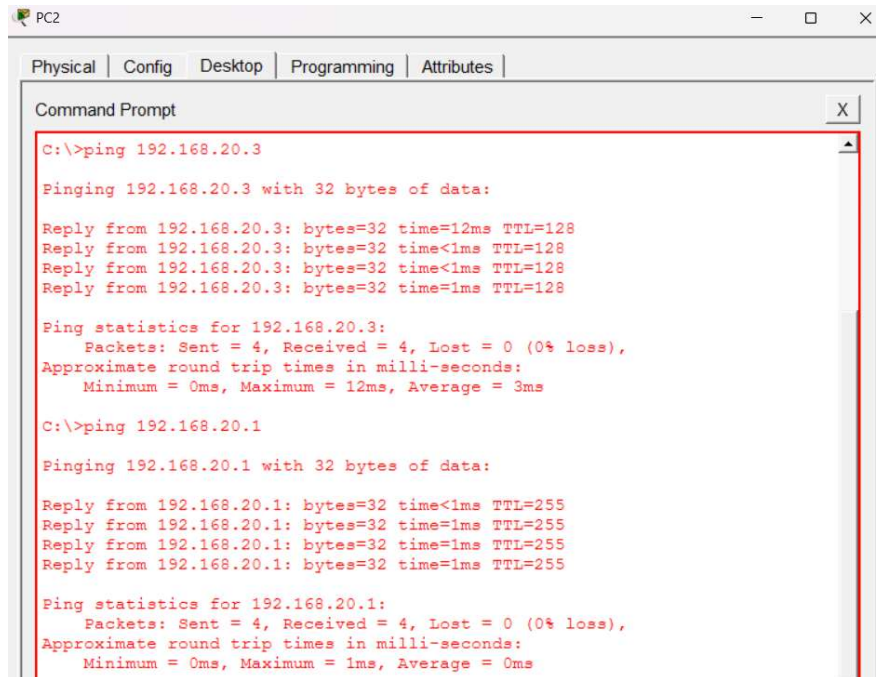
Symptom

Marketing cannot reach Engineering's file server, although pings inside its own VLAN work.

Method: the ping ladder

The ping ladder tests from the closest device outward. The first step that fails shows where the fault begins. All tests below were run from PC2 in Marketing (VLAN 10).

1. ping 192.168.20.3 (Marketing PC) - 4/4 replies, TTL 128 – proves layer 2 inside VLAN 10 works.
2. ping 192.168.20.1 (Marketing gateway) - 4/4 replies, TTL 255 – proves trunk and sub-interface Gi0/0.10 work. TTL 255 means the reply came from the router itself.
3. ping 192.168.20.17 (Engineering gateway) - 4/4 replies, TTL 255 – Marketing reaches the router, and Gi0/0.20 is up with the right IP.
4. ping 192.168.20.19 (file server) - 0/4 replies, 100% loss - The fault lies beyond the router, on the Engineering side.



```
PC2
Physical | Config | Desktop | Programming | Attributes |
Command Prompt
C:\>ping 192.168.20.3

Pinging 192.168.20.3 with 32 bytes of data:

Reply from 192.168.20.3: bytes=32 time=12ms TTL=128
Reply from 192.168.20.3: bytes=32 time<1ms TTL=128
Reply from 192.168.20.3: bytes=32 time<1ms TTL=128
Reply from 192.168.20.3: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.20.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 3ms

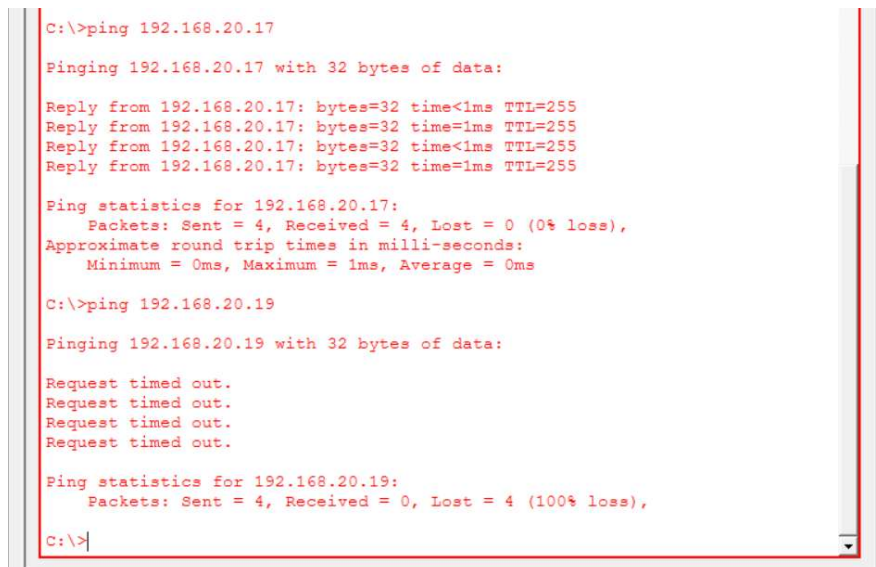
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time=1ms TTL=255
Reply from 192.168.20.1: bytes=32 time=1ms TTL=255
Reply from 192.168.20.1: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Figure 5: PC2 pings the Marketing PC (192.168.20.3) and the Marketing gateway (192.168.20.1)



```
C:\>ping 192.168.20.17

Pinging 192.168.20.17 with 32 bytes of data:

Reply from 192.168.20.17: bytes=32 time<1ms TTL=255
Reply from 192.168.20.17: bytes=32 time=1ms TTL=255
Reply from 192.168.20.17: bytes=32 time<1ms TTL=255
Reply from 192.168.20.17: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.20.17:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.20.19

Pinging 192.168.20.19 with 32 bytes of data:

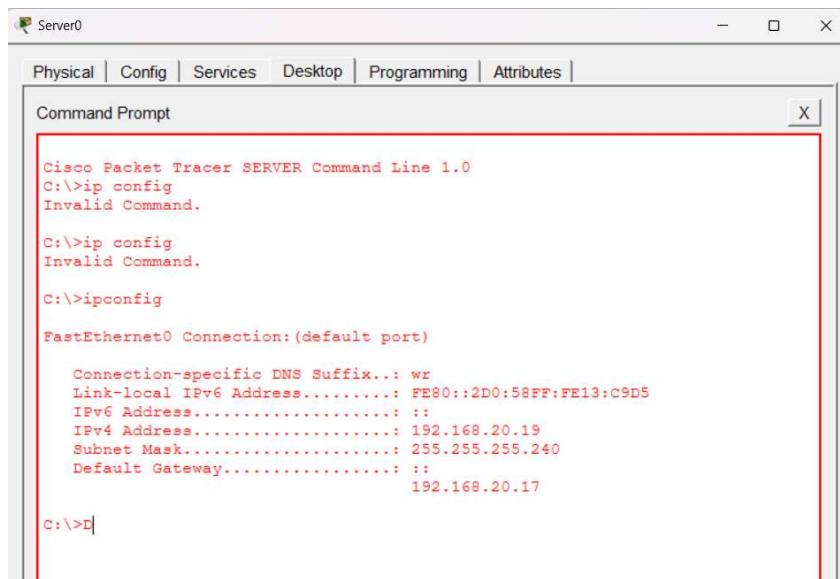
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.19:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Figure 6: PC2 pings the Engineering gateway (192.168.20.17) - success - and the server (192.168.20.19) - 100% loss

Testing from the server



```
Server0
Physical | Config | Services | Desktop | Programming | Attributes |
Command Prompt
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ip config
Invalid Command.

C:\>ip config
Invalid Command.

C:\>ipconfig

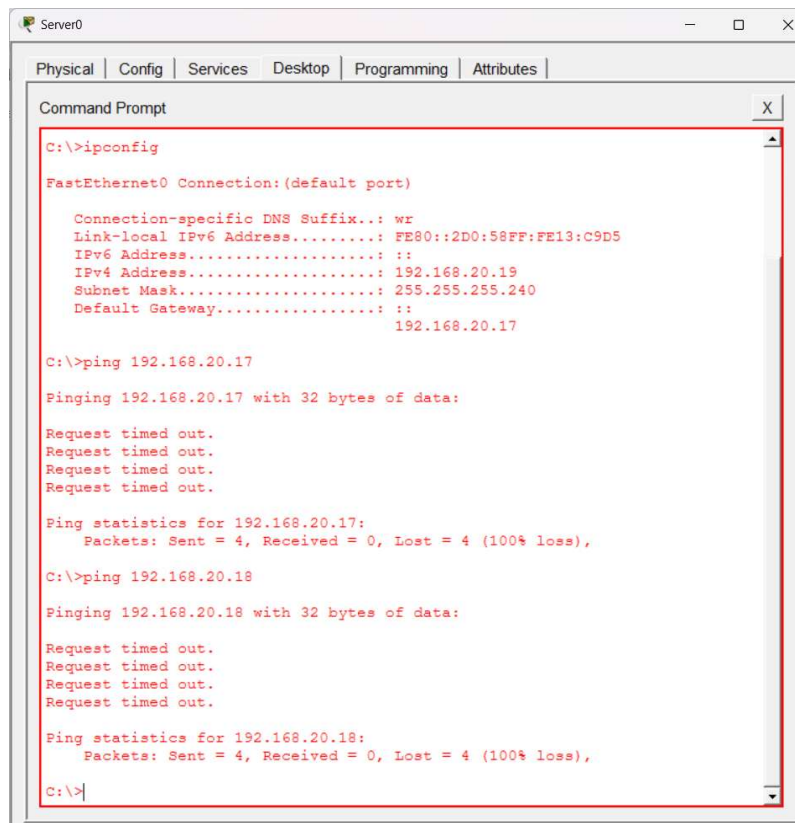
FastEthernet0 Connection: (default port)

Connection-specific DNS Suffix...: wr
Link-local IPv6 Address...: FE80::2D0:58FF:FE13:C9D5
IPv6 Address...: ::
IPv4 Address...: 192.168.20.19
Subnet Mask...: 255.255.255.240
Default Gateway...: ::
                  192.168.20.17

C:\>|
```

Figure 7: Server0 ipconfig - IP 192.168.20.19, mask 255.255.255.240, gateway 192.168.20.17

IPCONFIG command was used to verify, the server's IP address, subnet mask and default gateway are all correct, so Layer 3 is ruled out.



```
Server0
Physical | Config | Services | Desktop | Programming | Attributes |
Command Prompt
C:\>ipconfig

FastEthernet0 Connection: (default port)

Connection-specific DNS Suffix...: wr
Link-local IPv6 Address...: FE80::2D0:58FF:FE13:C9D5
IPv6 Address...: ::
IPv4 Address...: 192.168.20.19
Subnet Mask...: 255.255.255.240
Default Gateway...: ::
                  192.168.20.17

C:\>ping 192.168.20.17

Pinging 192.168.20.17 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.17:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.20.18

Pinging 192.168.20.18 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.18:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>|
```

Figure 8: Server0 cannot ping its own gateway (192.168.20.17) or another Engineering host (192.168.20.18)

With correct addressing, the server cannot reach even its own gateway or another host in its VLAN. Nothing is leaving the server, which points to the link between the server and the switch: a Layer 2 fault.

Locating the fault on the switch

```

Switch1
Physical | Config | CLI | Attributes |
IOS Command Line Interface

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/13, changed state to down

Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show interfaces status
Port      Name      Status      Vlan      Duplex  Speed  Type
Fa0/1     Fa0/1     connected   10        a-full  a-100  10/100BaseTX
Fa0/2     Fa0/2     notconnect  10        auto    auto   10/100BaseTX
Fa0/3     Fa0/3     connected   10        a-full  a-100  10/100BaseTX
Fa0/4     Fa0/4     notconnect  10        auto    auto   10/100BaseTX
Fa0/5     Fa0/5     notconnect  10        auto    auto   10/100BaseTX
Fa0/6     Fa0/6     notconnect  10        auto    auto   10/100BaseTX
Fa0/7     Fa0/7     notconnect  10        auto    auto   10/100BaseTX
Fa0/8     Fa0/8     notconnect  10        auto    auto   10/100BaseTX
Fa0/9     Fa0/9     notconnect  10        auto    auto   10/100BaseTX
Fa0/10    Fa0/10    notconnect  10        auto    auto   10/100BaseTX
Fa0/11    Fa0/11    connected   20        a-full  a-100  10/100BaseTX
Fa0/12    Fa0/12    notconnect  20        auto    auto   10/100BaseTX
Fa0/13    Fa0/13    disabled    20        auto    auto   10/100BaseTX
Fa0/14    Fa0/14    notconnect  20        auto    auto   10/100BaseTX
Fa0/15    Fa0/15    notconnect  20        auto    auto   10/100BaseTX
Fa0/16    Fa0/16    notconnect  20        auto    auto   10/100BaseTX
Fa0/17    Fa0/17    notconnect  20        auto    auto   10/100BaseTX
Fa0/18    Fa0/18    notconnect  20        auto    auto   10/100BaseTX
Fa0/19    Fa0/19    notconnect  20        auto    auto   10/100BaseTX
Fa0/20    Fa0/20    notconnect  20        auto    auto   10/100BaseTX
Fa0/21    Fa0/21    notconnect  1         auto    auto   10/100BaseTX
Fa0/22    Fa0/22    notconnect  1         auto    auto   10/100BaseTX
Fa0/23    Fa0/23    notconnect  1         auto    auto   10/100BaseTX
Fa0/24    Fa0/24    connected   trunk     a-full  a-100  10/100BaseTX
Gig0/1    Gig0/1    notconnect  1         auto    auto   10/100/1000BaseTX
Gig0/2    Gig0/2    notconnect  1         auto    auto   10/100/1000BaseTX

Switch#

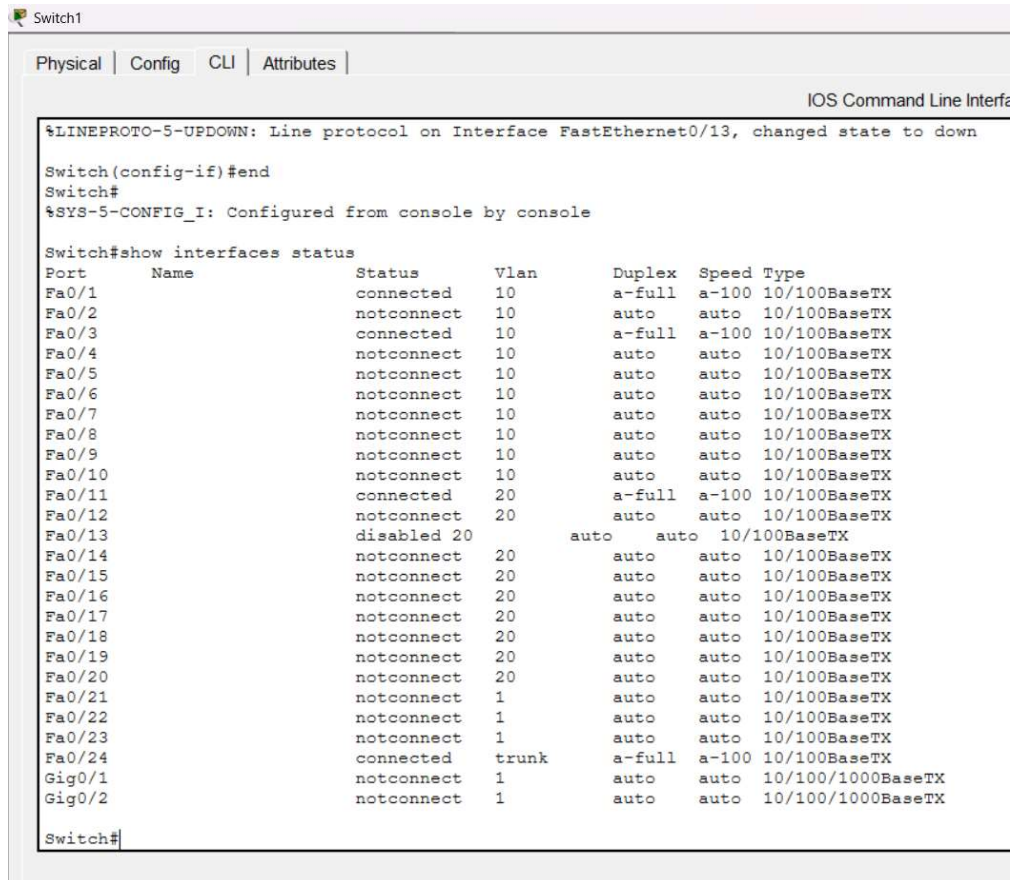
```

Figure 9: Switch1 "show interfaces status" - Fa0/13 shows "disabled" in VLAN 20

Fa0/13 is in the correct VLAN (20), but its status is disabled. Every other Engineering port shows either connected or notconnect.

Port status	Meaning
disabled	The port was shut down by an administrator (the "shutdown" command). This is a configuration fault.
notconnect	The port is enabled but no link is detected (cable unplugged or device off).
connected	The port is enabled and a link is up.

5.4 Locating the fault on the switch



```
Switch1
Physical | Config | CLI | Attributes |
IOS Command Line Interface

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/13, changed state to down

Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show interfaces status
Port      Name      Status      Vlan      Duplex  Speed  Type
-----
Fa0/1     connected 10          a-full    a-100   10/100BaseTX
Fa0/2     notconnect 10          auto      auto    10/100BaseTX
Fa0/3     connected 10          a-full    a-100   10/100BaseTX
Fa0/4     notconnect 10          auto      auto    10/100BaseTX
Fa0/5     notconnect 10          auto      auto    10/100BaseTX
Fa0/6     notconnect 10          auto      auto    10/100BaseTX
Fa0/7     notconnect 10          auto      auto    10/100BaseTX
Fa0/8     notconnect 10          auto      auto    10/100BaseTX
Fa0/9     notconnect 10          auto      auto    10/100BaseTX
Fa0/10    notconnect 10          auto      auto    10/100BaseTX
Fa0/11    connected 20          a-full    a-100   10/100BaseTX
Fa0/12    notconnect 20          auto      auto    10/100BaseTX
Fa0/13    disabled 20          auto      auto    10/100BaseTX
Fa0/14    notconnect 20          auto      auto    10/100BaseTX
Fa0/15    notconnect 20          auto      auto    10/100BaseTX
Fa0/16    notconnect 20          auto      auto    10/100BaseTX
Fa0/17    notconnect 20          auto      auto    10/100BaseTX
Fa0/18    notconnect 20          auto      auto    10/100BaseTX
Fa0/19    notconnect 20          auto      auto    10/100BaseTX
Fa0/20    notconnect 20          auto      auto    10/100BaseTX
Fa0/21    notconnect 1           auto      auto    10/100BaseTX
Fa0/22    notconnect 1           auto      auto    10/100BaseTX
Fa0/23    notconnect 1           auto      auto    10/100BaseTX
Fa0/24    connected trunk       a-full    a-100   10/100BaseTX
Gig0/1    notconnect 1           auto      auto    10/100/1000BaseTX
Gig0/2    notconnect 1           auto      auto    10/100/1000BaseTX

Switch#
```

Figure 9: Switch1 "show interfaces status" - Fa0/13 shows "disabled" in VLAN 20

Fa0/13 is in the correct VLAN (20), but its status is disabled. Every other Engineering port shows either connected or notconnect.

Port status	Meaning
disabled	The port was shut down by an administrator (the "shutdown" command). This is a configuration fault.
notconnect	The port is enabled but no link is detected (cable unplugged or device off).
connected	The port is enabled and a link is up.

Root Cause

Layer 2 fault: switch port FastEthernet0/13 (VLAN 20) is administratively shut down. The Engineering file server cannot send or receive frames, so it is unreachable from Marketing and from the rest of its own VLAN. The IP addressing, the trunk and the router sub-interfaces are all correct.

Layer	Verdict and evidence
Layer 3	Ruled out. The server's IP, mask and gateway are correct (Figure 7), and the router sub-interface answers pings (Figure 6).
Layer 2	Confirmed. The server cannot reach its own gateway (Figure 8), and Fa0/13 is disabled (Figure 9).

FIX:

The port fa0/13 was enabled by administrator using command – “No shut” inside interface fa0/13.

```
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int fa0/13
Switch(config-if)#no shut

Switch(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/13, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/13, changed
state to up

Switch(config-if)#
```

Figure 10: Switch1 - Fa0/13 enabled in VLAN 20

Verification using ping was successful.

```
C:\>ping 192.168.20.19

Pinging 192.168.20.19 with 32 bytes of data:

Reply from 192.168.20.19: bytes=32 time<1ms TTL=127
Reply from 192.168.20.19: bytes=32 time=1ms TTL=127
Reply from 192.168.20.19: bytes=32 time=1ms TTL=127
Reply from 192.168.20.19: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.19:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Figure 10:PC2 (Marketing): ping 192.168.20.19 succeed with TTL=127

D Section D — AI-Augmented Learning

STEP 1 · BUILD WITH AI

Use an AI tool of your choice (ChatGPT, Claude, GitHub Copilot, etc.) to help you generate a complete Cisco IOS configuration script that:

- Configures VLAN 10 and VLAN 20 with port assignments on the switch.
- Configures an 802.1Q trunk link between the switch and router.
- Configures Router-on-a-Stick sub-interfaces on the router for inter-VLAN routing.
- Demonstrates the working configuration by pinging successfully between a VLAN 10 host and a VLAN 20 host in Packet Tracer.

STEP 2 · TEST & DEBUG (WITHOUT AI)

Then, working without AI, test the configuration in Packet Tracer and find at least one bug, missing command, or misconfiguration in the AI's solution (e.g. a missing 'no shutdown' on a sub-interface, wrong encapsulation VLAN ID, or a trunk not allowing both VLANs). Fix it yourself.

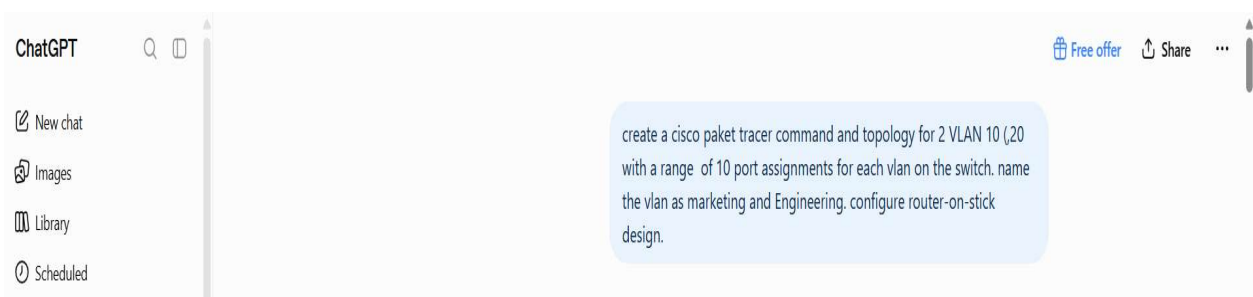
Answer:

SUBMIT — All 3 items required

- The exact prompt(s) you gave the AI tool.
- The AI's original configuration script and your corrected version.
- A 3–4 line note explaining what you changed and why the AI's version needed it.

Answer:

- The exact prompt(s) i gave the AI tool.



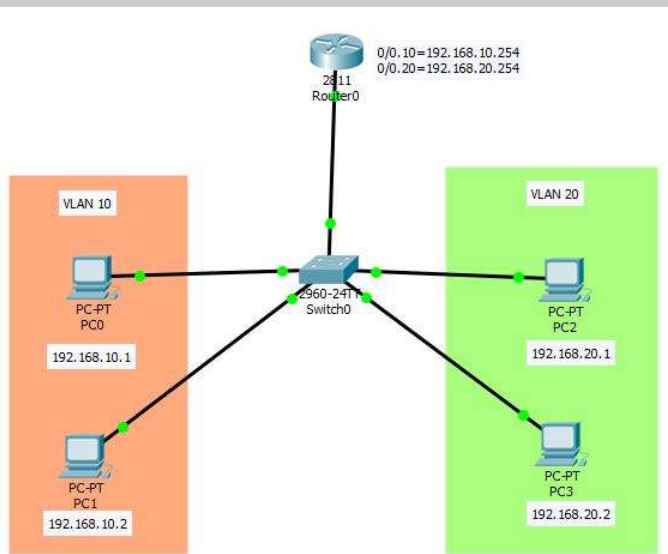
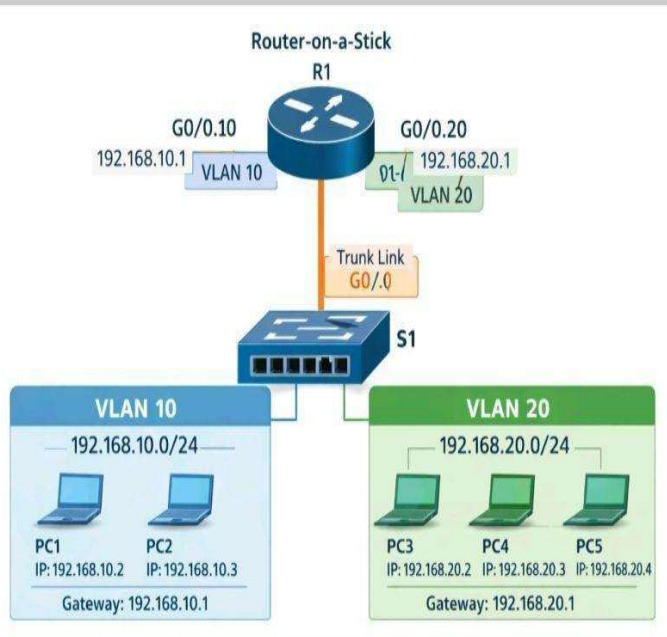
- The AI's original configuration script and your corrected version.

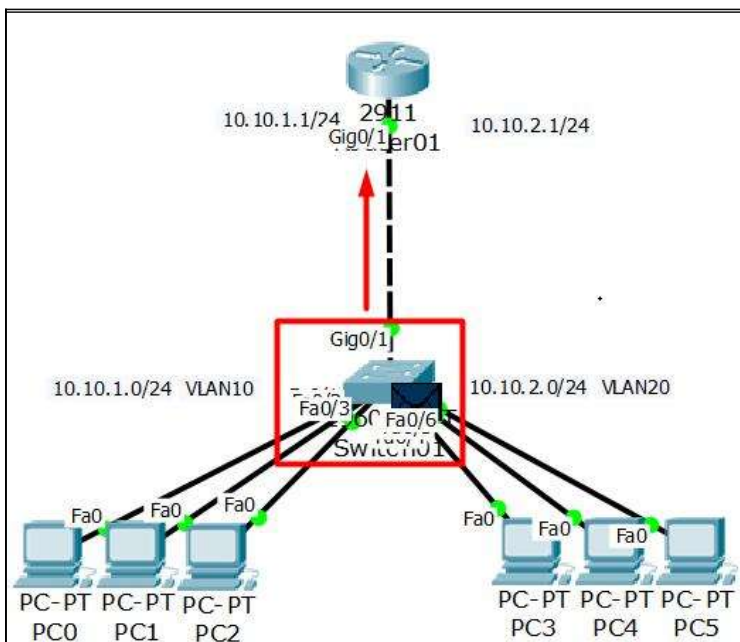
Yes. Below is a Cisco Packet Tracer Router-on-a-Stick topology with:

- VLAN 10 – Marketing → 10 switch ports

- VLAN 20 – Engineering → 10 switch ports
- 20 PCs total
- 1 Cisco 2960 switch
- 1 Cisco router
- Router-on-a-Stick for communication between VLANs
- Trunk between router and switch

1. Topology





2. IP Addressing Plan

VLAN	Name	Ports	Network	Gateway
VLAN 10	Marketing	Fa0/1–Fa0/10	192.168.10.0/24	192.168.10.1
VLAN 20	Engineering	Fa0/11–Fa0/20	192.168.20.0/24	192.168.20.1

Example PC addresses

Marketing PCs

PC1 → 192.168.10.11
 PC2 → 192.168.10.12
 PC3 → 192.168.10.13
 ...
 PC10 → 192.168.10.20

Gateway:

192.168.10.1

Engineering PCs

PC11 → 192.168.20.11
 PC12 → 192.168.20.12
 PC13 → 192.168.20.13
 ...
 PC20 → 192.168.20.20

Gateway:

192.168.20.1

3. Switch Configuration

Connect:

Router G0/0 ————— Switch Fa0/24

Then enter the following on SW1:

```
enable
configure terminal
```

```
hostname SW1
```

```
! Create VLAN 10
vlan 10
name Marketing
exit
```

```
! Create VLAN 20
vlan 20
name Engineering
exit
```

```
! Assign ports Fa0/1 to Fa0/10 to Marketing
interface range fastEthernet 0/1-10
switchport mode access
switchport access vlan 10
spanning-tree portfast
exit
```

```
! Assign ports Fa0/11 to Fa0/20 to Engineering
interface range fastEthernet 0/11-20
switchport mode access
switchport access vlan 20
spanning-tree portfast
exit
```

```
! Configure trunk to router
interface fastEthernet 0/24
switchport mode trunk
switchport trunk allowed vlan 10,20
exit
```

```
end
write memory
```

Important

The port assignment is:

Fa0/1 — Fa0/10 → VLAN 10 Marketing

Fa0/11 — Fa0/20 → VLAN 20 Engineering

Fa0/24 → TRUNK to Router

4. Router Configuration – Router-on-a-Stick

On R1, configure the physical interface and two subinterfaces.

enable

configure terminal

hostname R1

! Enable physical interface

interface gigabitEthernet 0/0

no shutdown

exit

! VLAN 10 subinterface

interface gigabitEthernet 0/0.10

encapsulation dot1Q 10

ip address 192.168.10.1 255.255.255.0

exit

! VLAN 20 subinterface

interface gigabitEthernet 0/0.20

encapsulation dot1Q 20

ip address 192.168.20.1 255.255.255.0

exit

end

write memory

The router now has:

G0/0.10

VLAN 10

192.168.10.1

G0/0.20

VLAN 20

192.168.20.1

These subinterfaces act as the default gateways for the two VLANs.

5. Configure the PCs

For each PC in Packet Tracer:

PC → Desktop → IP Configuration

Marketing PC

For example, PC1:

IP Address: 192.168.10.11
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.10.1

PC2:

IP Address: 192.168.10.12
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.10.1

Continue up to PC10.

Engineering PC

PC11:

IP Address: 192.168.20.11
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.20.1

PC12:

IP Address: 192.168.20.12
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.20.1

Continue up to PC20.

6. Verify the VLANs

On the switch:

show vlan brief

You should see approximately:

VLAN Name	Status	Ports
10 Marketing	active	Fa0/1, Fa0/2, ... Fa0/10
20 Engineering	active	Fa0/11, Fa0/12, ... Fa0/20

Check the trunk:

show interfaces trunk

You should see:

Port	Mode	Encapsulation	Status
Fa0/24	on	802.1q	trunking

7. Verify Router Subinterfaces

On R1:

show ip interface brief

Expected result:

Interface	IP-Address	Status
GigabitEthernet0/0	unassigned	up
GigabitEthernet0/0.10	192.168.10.1	up
GigabitEthernet0/0.20	192.168.20.1	up

8. Test the Configuration

From a Marketing PC, open Command Prompt:

ping 192.168.10.1

This tests the Marketing gateway.

Then:

ping 192.168.20.1

This tests connectivity to the Engineering gateway.

Finally, from Marketing PC1:

ping 192.168.20.11

If successful, you have demonstrated inter-VLAN routing.

How the traffic works

Marketing PC

192.168.10.11



SW1 Fa0/1

VLAN 10



Fa0/24 TRUNK



R1 G0/0.10

192.168.10.1



Router performs routing

R1 G0/0.20

192.168.20.1



TRUNK



▼
SW1
VLAN 20
↓
▼
Engineering PC
192.168.20.11

This is the classic Router-on-a-Stick design: one physical router interface carries multiple VLANs using 802.1Q tagging, with one router subinterface per VLAN.

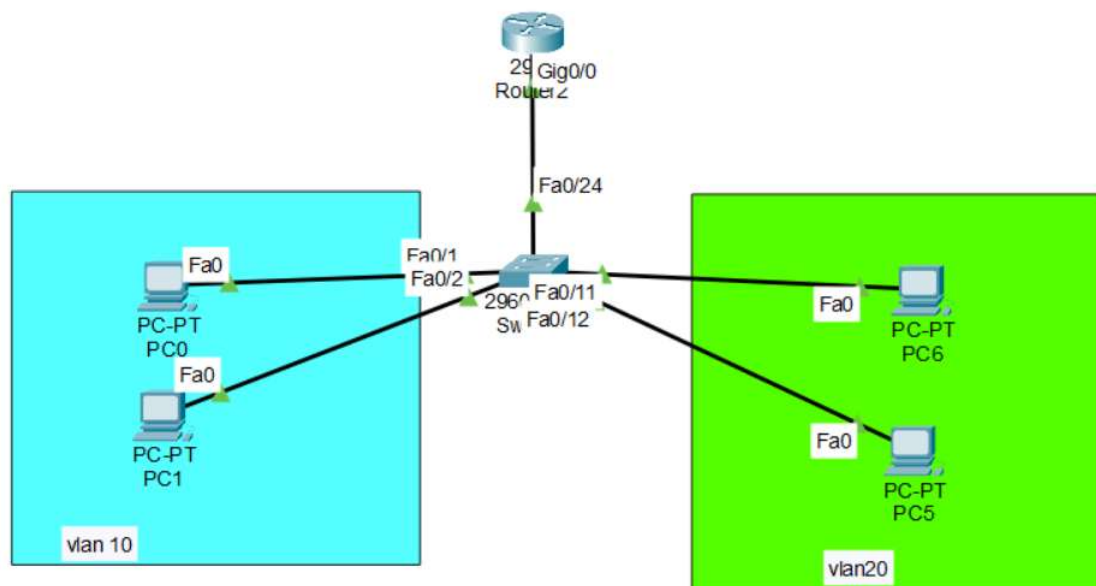


Figure a. The topology created

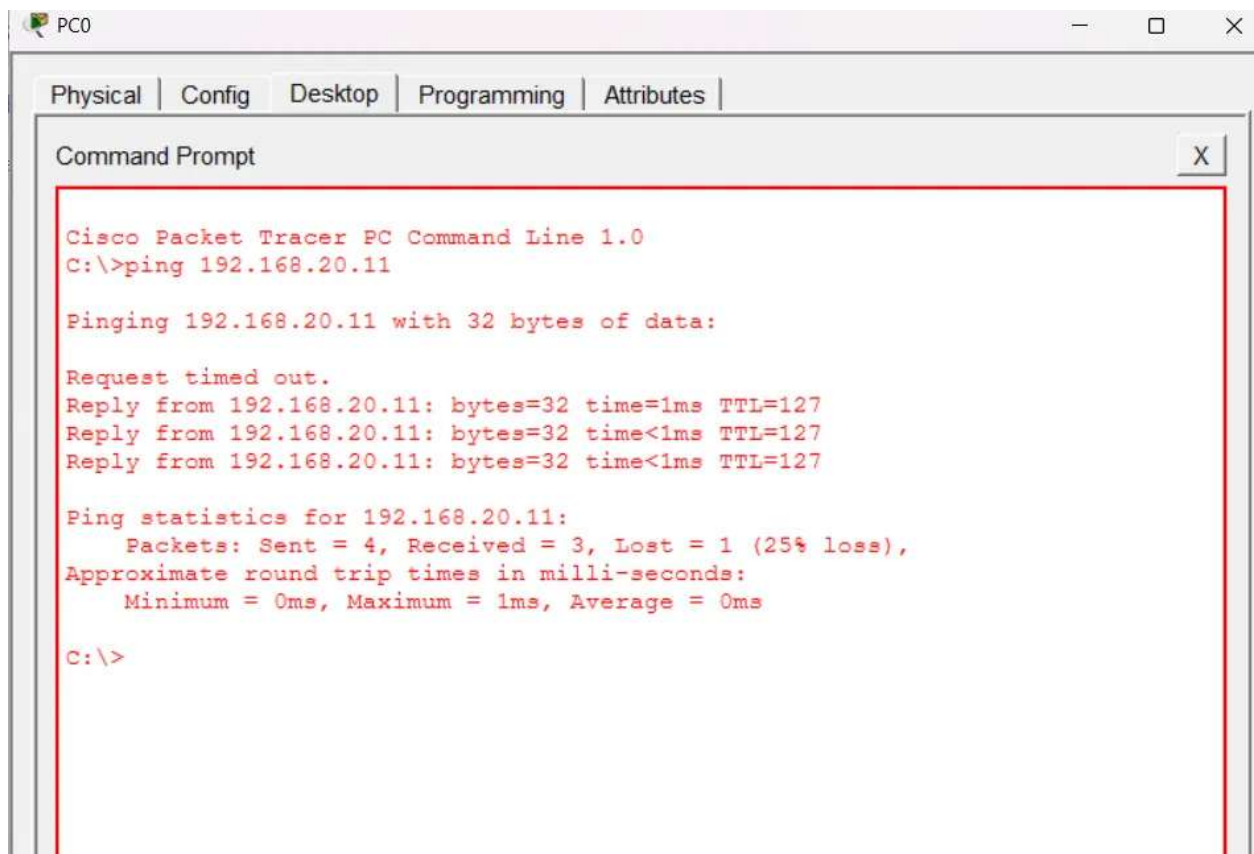


Figure b. Successful ping with TTL=127

The AI's script worked: PC0 in VLAN 10 pinged a VLAN 20 host with TTL=127. But as per ChatGPT script PortFast was enabled on all ports Fa0/1-20 without BPDU Guard, and Packet Tracer warned about this. I added spanning-tree bpduguard enable to both access port ranges. When I connected a second switch to Fa0/3, the port went err-disabled, so the script needed this safeguard to protect its access ports.

The command added:

```
interface range fastEthernet 0/1 - 10
spanning-tree bpduguard enable
interface range fastEthernet 0/11 - 20
spanning-tree bpduguard enable
```

```
%Portfast has been configured on FastEthernet0/10 but will only
have effect when the interface is in a non-trunking mode.
SW1(config-if-range)#exit
SW1(config)#interface fastEthernet 0/24
SW1(config-if)#switchport mode trunk
SW1(config-if)#switchport trunk allowed vlan 10,20
SW1(config-if)#exit
SW1(config)#interface range fastEthernet 0/11-20
SW1(config-if-range)#switchport mode access
SW1(config-if-range)#switchport access vlan 20
SW1(config-if-range)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/11 but will only
have effect when the interface is in a non-trunking mode.
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/12 but will only
have effect when the interface is in a non-trunking mode.
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION
```

Figure c: The portfast error warning

What is PORTFAST:

When we plug a device into a switch port, spanning tree (STP) doesn't let traffic through immediately. It will first check that the new link won't create a loop. Each port steps through **blocking, listening, learning, then forwarding**, and that takes roughly 30 seconds which can cause delay in DHCP configuration of a PC or server.

In order to avoid this delay and we are sure that only end devices are going to be connected to the ports, we use the command "spanning-tree portfast" – this will immediately start forwarding packets through. PortFast only skips the 30-second startup delay

But without a safeguard, when a switch or hub is accidentally connected to such a port it will create a loop and before spanning tree notices, and cause temporary bridging loops.

PortFast and BPDU Guard are normally used as a pair:

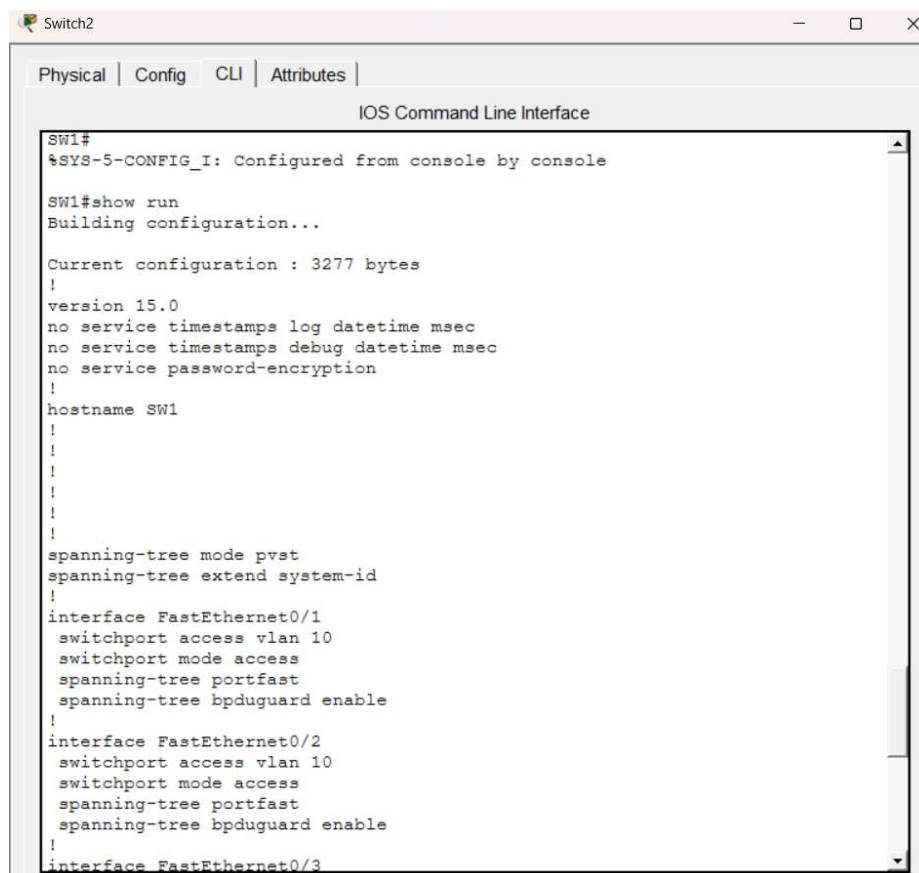
- PortFast makes the port fast.
- BPDU Guard makes it safe.

What BPDU Guard does

Switches talk to each other using small messages called **BPDU**s (Bridge Protocol Data Units). PCs, printers and servers never send them. So if a BPDU arrives on an access port, a switch or other network device is plugged in where it shouldn't be.

BPDU Guard reacts to that by shutting the port down. It puts the port into **err-disabled** state, which stops the loop risk before it spreads. To recover, you fix the problem, then run `shutdown` followed by `no shutdown` on the port.

The fix:

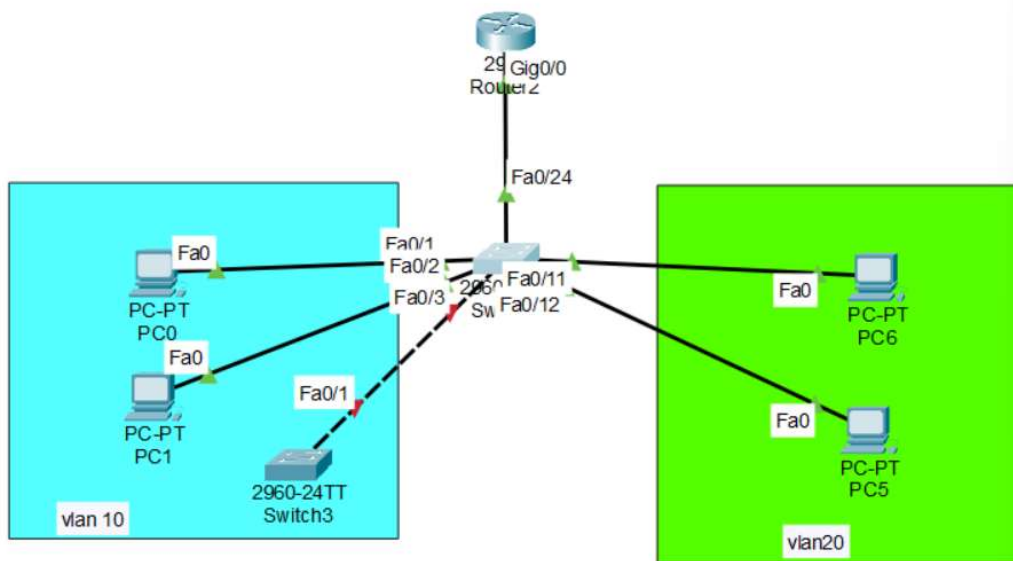


```
Switch2
Physical | Config | CLI | Attributes |
IOS Command Line Interface
SW1#
%SYS-5-CONFIG_I: Configured from console by console

SW1#show run
Building configuration...

Current configuration : 3277 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname SW1
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport access vlan 10
 switchport mode access
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/2
 switchport access vlan 10
 switchport mode access
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/3
```

Checking if the safeguard is working, an additional switch is connected to the interface fa0/3 and it was observed BPDU safeguard activated after detecting a switch is connected and disabled the port.



Switch2

Physical | Config | CLI | Attributes

IOS Command Line Interface

```

interface FastEthernet0/13
switchport access vlan 20
switchport mode access
spanning-tree portfast
spanning-tree bpduguard enable
!
interface FastEthernet0/14
switchport access vlan 20
switchport mode access
spanning-tree portfast
spanning-tree bpduguard enable
!
interface FastEthernet0/15
switchport access vlan 20
switchport mode access
spanning-tree portfast
spanning-tree bpduguard enable
!
interface FastEthernet0/16
--More--
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on
FastEthernet0/3 (10), with Switch FastEthernet0/1 (1).
%SPANTREE-2-BLOCK_BPDUGUARD: Received BPDU on port FastEthernet0/3 with BPDU
Guard enabled. Disabling port.

%PM-4-ERR_DISABLE: bpduguard error detected on 0/3, putting 0/3 in err-
disable state

%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to
administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed
state to down

```